

Actors, Use Cases, Architecture and Design Specification

Online Examination Portal

Team E6

1. Introduction

1.1 Purpose

This document defines the actors and use cases assigned to the team and extends them into the Phase-1 architecture and detailed design required for implementation.

1.2 Scope

The document covers actors, grouped use cases, use-case responsibilities, logical architecture, component descriptions, architecture traceability, security architecture, sequence diagrams, API design, error handling, data design, and deployment considerations.

1.3 Design Basis

The requirements and use-case IDs are based on the finalized Phase-1 handoff. Technology choices and API paths in this document are proposed design decisions and are not claimed as finalized implementation choices.

2. Actors and Responsibilities

Actor	Type	Responsibilities
Student	Human	Log in, view available examinations, start/attempt examinations, save/change answers, submit, view released results.
Teacher	Human	Log in, upload/manage questions, create/configure examinations, view examination reports.
Administrator	Human	Manage users, roles, access, and authorized system settings.
Authentication Service	System service	Validate credentials and return authentication outcome.
Examination Service	System service	Select and persist the examination question set and support examination initialization.
Evaluation Service	System service	Evaluate objective answers and calculate scores.
Result Service	System service	Release results according to configured release time.

3. Grouped Use Cases

Actor	ID	Use Case	Description
Student	UC-S01	Log In	Authenticate using valid credentials.

Student	UC-S02	View Available Examinations	View examinations available according to configured availability.
Student	UC-S03	Start Examination	Initialize an attempt and receive the selected question set.
Student	UC-S04	View Examination Questions	Display the questions associated with the active attempt in shuffled order.
Student	UC-S05	Save / Change Answers	Select, save, revisit, and change answers before submission.
Student	UC-S06	Submit Examination	Submit manually or be auto-submitted when time expires.
Student	UC-S07	View Released Result	View the result after the configured release time.
Teacher	UC-T01	Teacher Login	Authenticate as a teacher.
Teacher	UC-T02	Upload Questions	Add objective questions to the question bank.
Teacher	UC-T03	Assign Difficulty	Assign a difficulty level to a question.
Teacher	UC-T04	Edit / Remove Questions	Edit or remove questions before publication.
Teacher	UC-T05	Create Examination	Create an examination using configured settings.
Teacher	UC-T06	Configure Examination	Set duration and availability period.
Teacher	UC-T07	View Examination Reports	View reports for authorized examinations.
Administrator	UC-A01	Manage Users	Create/update/deactivate authorized user accounts.
Administrator	UC-A02	Manage Roles	Assign or manage system roles according to authorization.
Administrator	UC-A03	Manage System Settings	Maintain authorized system-level settings.
Administrator	UC-A04	Review Examination Information	Access authorized administrative examination information.
Administrator	UC-A05	Manage Access	Control administrative access and permissions.
Authentication Service	UC-X01	Authenticate User	Validate supplied credentials and return authentication outcome.
Examination Service	UC-X02	Select Question Set	Randomly select the configured number of questions and persist the selected set for the examination.
Evaluation Service	UC-X03	Evaluate Attempt	Evaluate objective answers and calculate the score.
Result Service	UC-X04	Release Result	Release the result automatically at the configured result-release time.
Administrator	UC-A06	Administrator Login	Authenticate as an administrator before accessing administrative functions.

4. Use Case Diagram

Updated in a separate file

5. Architecture

5.1 Architectural Pattern

Proposed pattern: layered architecture with service-oriented application modules. Presentation components communicate with application services; domain logic enforces examination rules; persistence manages system data; supporting services handle authentication, evaluation, and result release.

5.2 Component Diagram

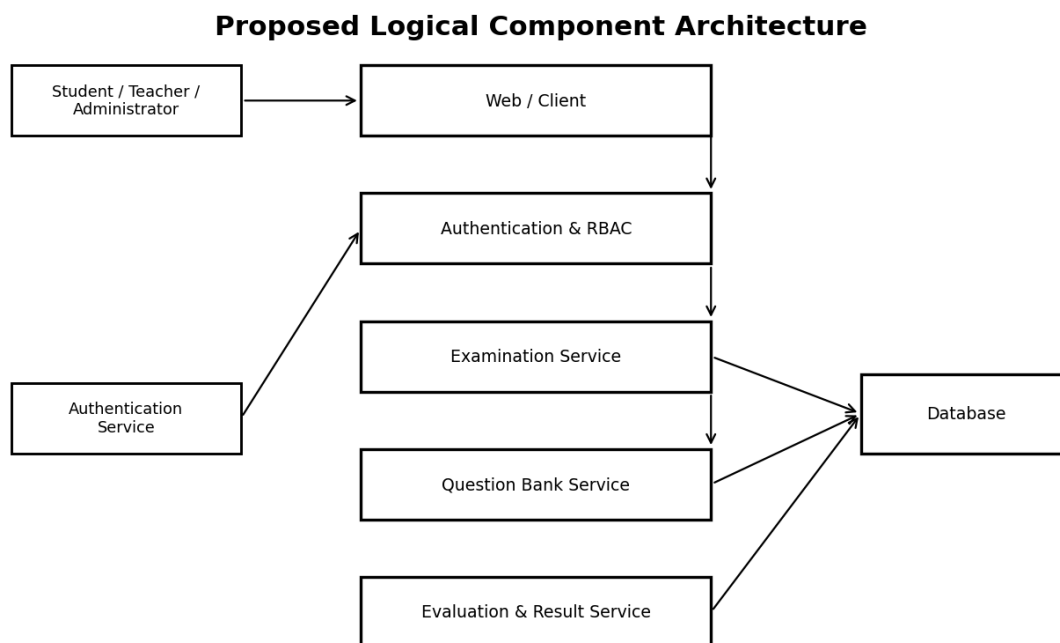


Figure 2. Proposed logical component architecture

5.3 Component Descriptions

Component	Responsibility
Web / UI Layer	Login, question management, examination screens, result/report screens, administration screens.
Authentication Module	Credential validation, session creation and expiry, role retrieval.
User & Role Module	User management and role/permission administration.
Question Bank Module	Question upload, difficulty assignment, pre-publication edit/remove operations.
Examination Module	Create/configure examinations, availability, duration, attempt initialization.
Attempt Module	Display active questions, save/change answers, manual and automatic submission.
Evaluation Module	Evaluate objective answers and calculate score.
Result Module	Store and release results at configured time.
Reporting Module	Provide authorized teacher examination reports.

Database	Persistent storage for users, roles, questions, examinations, attempts, answers, scores, results, and audit records.
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5.4 Architecture Traceability

Architecture Area	Requirements
Authentication	FR-01, FR-02, SEC-01, SEC-02, FR-22
User and Role Management	FR-03, SEC-02, SEC-03
Question Bank	FR-05, FR-06, FR-07
Examination Management	FR-04, FR-08, FR-21
Question Selection / Ordering	FR-09, FR-10, FR-11
Attempt Management	FR-12, FR-13, FR-14, FR-15, FR-18
Evaluation	FR-16, FR-17
Result Management	FR-19
Reporting	FR-20
Quality / Security	NFR-01 to NFR-12, SEC-01 to SEC-08

5.5 Security Architecture

- Enforce authorization at service/API boundaries rather than relying only on the user interface.
- Protect unreleased questions, active attempts, and results according to examination state and release time.
- Treat submitted attempts and answers as protected records after submission.
- Validate external input before processing or persistence.
- Centralize session lifecycle and expiry.
- Record security-relevant administrative actions in protected audit logs.
- Do not expose credentials or secrets to the client.

5.6 Performance and Scalability

- Persist the selected question set when an examination attempt is initialized so every student receives the same selected set.
- Keep common request paths lightweight and avoid unnecessary database round trips.
- Use transaction boundaries around answer submission, final submission, and evaluation.
- Measure the proposed 3-second response target using VAL-N03 after approval.
- Evaluate concurrent examination behavior using VAL-N10.

6. Detailed Design

6.1 Sequence Diagram : Authentication

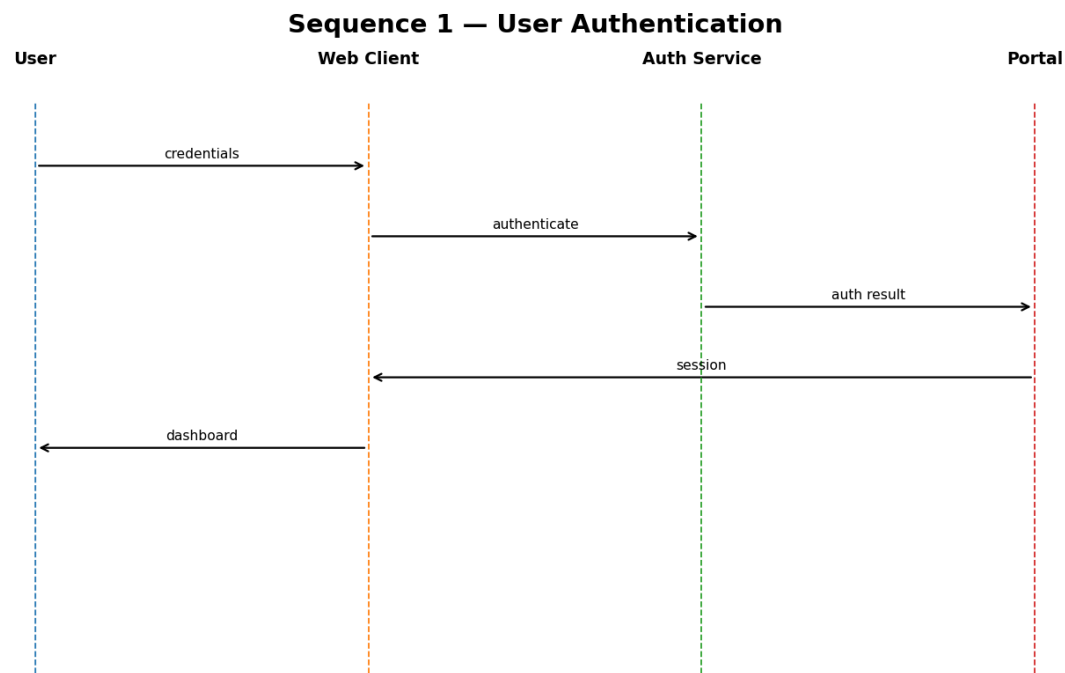


Figure 3. Authentication sequence

6.2 Sequence Diagram : Examination Lifecycle

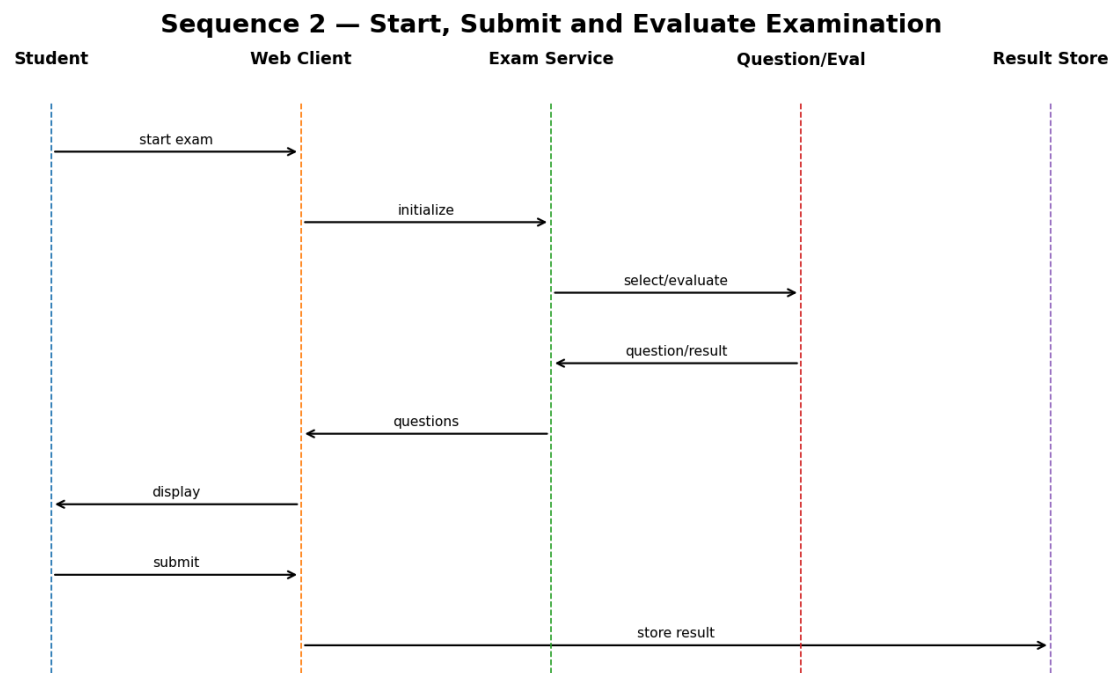


Figure 4. Examination lifecycle sequence

6.3 API Design

Method	Proposed Endpoint	Purpose	Primary Requirements
POST	/api/auth/login	Authenticate a user	FR-01, FR-02, SEC-01
GET	/api/exams/available	List examinations currently available to a student	FR-21
POST	/api/exams	Create an examination	FR-04
POST	/api/questions	Upload a question	FR-05
PATCH	/api/questions/{id}	Edit a question before publication	FR-07
DELETE	/api/questions/{id}	Remove a question before publication	FR-07
POST	/api/exams/{id}/start	Initialize an examination attempt	FR-09, FR-10, FR-11, FR-12
PATCH	/api/attempts/{id}/answers	Save/change an answer	FR-13
POST	/api/attempts/{id}/submit	Submit an examination	FR-15
POST	/api/attempts/{id}/auto-submit	Auto-submit at timeout	FR-14
GET	/api/results/{attemptId}	Retrieve a released result	FR-19
GET	/api/exams/{id}/reports	View an authorized examination report	FR-20

The API paths above are proposed interfaces for design purposes and can be changed during implementation.

6.4 Error Handling

Code	Meaning	Example
200	Successful request	Result retrieved successfully.
201	Resource created	Question or examination created.
400	Invalid request	Required field missing or malformed.
401	Unauthenticated	Login required or session invalid.
403	Forbidden	User lacks the required role/permission.
404	Not found	Requested examination or question does not exist.
409	Conflict	Attempt already submitted or question locked after publication.
422	Validation error	Input fails a defined business rule.
500	Internal error	Unexpected server-side failure.

- Return a stable error code and human-readable message.
- Do not expose stack traces, credentials, or sensitive internal information to users.
- Log server-side diagnostic details separately from the client response.
- Keep error handling consistent across API modules.

6.5 Examination State Model

- Draft → Published → Available → Attempt Active → Submitted → Evaluated → Result Released.
- Question editing/removal is allowed only before the associated examination is published.
- Once submitted, an attempt is treated as protected examination data.
- Results become visible only after the configured release time.

6.6 Data Design

Entity	Key Data
User	user_id, credentials reference, role, status
Question	question_id, text, options, correct_answer, difficulty, publication state
Examination	exam_id, title, question_count, duration, availability, release_time, status
Attempt	attempt_id, exam_id, student_id, start_time, submit_time, status
Answer	attempt_id, question_id, selected_answer, saved_at

Result	attempt_id, score, evaluated_at, release_time, release_state
Audit Record	audit_id, actor_id, action, timestamp, target, outcome

7. Design Traceability to Validation

Design Area	Validation
Authentication and authorization	VAL-F01, VAL-S01 to VAL-S05
Question management	VAL-F02, VAL-F03, VAL-F13
Examination configuration	VAL-F04
Question selection and ordering	VAL-F05, VAL-F06
Answer handling and submission	VAL-F12, VAL-F07
Evaluation and score	VAL-F08
Result release	VAL-F09, VAL-F10
Reports	VAL-F14
Performance	VAL-N03, VAL-N10
Security	VAL-S01 to VAL-S11

8. Deployment and Operational Considerations

- Separate application and database responsibilities so data persistence is protected from presentation concerns.
- Use secure transport for authenticated and examination traffic.
- Back up examination and result data according to the final deployment policy.
- Monitor application errors and security-relevant audit events.
- Apply the same authorization rules in the UI and backend services.